

A Training-Free Quantum-Walk Scanner for Cryptic Allosteric Sites in Undruggable Proteins

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Abstract

Over 85% of disease-causing proteins are considered undruggable, and for these the only viable strategy is allostery. We propose a training-free scanner that takes an apo structure and blind-predicts distal regulatory pockets. Contact topology and evolutionary conservation are encoded as a sparse Hermitian Hamiltonian carrying a chiral phase, and signal propagation is the circuit e^{-iHt} acting on an active-site source state. The readout is the net *directed* current of that evolution: the difference between forward and reverse transport, an $N \times N$ directed connectivity matrix that is identically zero for any symmetric classical walk, so the signal it carries is quantum by construction and no diffusive measure on the same graph can express it. Residues are ranked by their directed coupling to the active site, deconfounded of both degree and proximity, which is the statistic the challenge scores. An active-site-anchored Krylov projection reduces the walk to a one-dimensional chain that preserves source-to-residue transport exactly, decoupling qubit count from protein length and bounding two-qubit depth independently of system size. Because no parameters are fitted, the targets serve purely as blind tests, and a Lindblad dephasing sweep places the classical diffusive baseline at the incoherent endpoint of the same model, where the directed signal vanishes.

1. Understanding of the Problem

The stated need is a scanner placed upstream of screening: it ingests static structures of targets already shelved as undruggable and returns a probability map of cryptic and allosteric sites, so library design can be aimed at regions mechanistically connected to disease function.

Three properties of the problem constrain any solution. The pockets are cryptic, so scanning for surface concavity misses them by construction, which is why we do not use pocket geometry as a filter. The signal is dynamic: a site is allosteric because perturbation there reaches the active site, so the method must model propagation. And molecular-dynamics trajectories are excluded, ruling out the methods that currently dominate cryptic-pocket discovery.

The elastic-network hypothesis makes the problem tractable. If contact topology drives collective dynamics, a protein reduces to a graph and atomic force fields can be set aside; the Gaussian Network Model reproduces crystallographic B-factors from a single-parameter C α contact network [10], and topology-driven signal propagation is well established [12]. A graph is also the natural setting for a quantum walk.

2. Prior Work and the Gap

Classical graph methods. Elastic network models [10, 11] predict fluctuations and collective modes from topology; Estrada communicability e^A [8] measures diffusive connectivity. The strongest published result in our exact setting is bond-to-bond propensity analysis, which recovers allosteric sites knowing only the orthosteric site [3]. These are fast, MD-free and the honest baselines; their common feature is that propagation is diffusive.

Pocket and learned predictors. fpocket scores geometric concavity but cannot see a pocket closed in the apo state. The state of the art for cryptic sites is PocketMiner, a graph neural network reaching ROC-AUC 0.87 on 39 confirmed cryptic pockets [13]. Its constraint is supervision: labelled apo-holo data is scarce (ASBench, a few hundred entries [14]; CASBench, 91 enzymes), limiting generalisation to novel targets. A training-free method sidesteps this.

Quantum approaches. Continuous-time quantum walks are mature, with a spectral theory of the average mixing matrix [1]; open-system generalisations interpolate between quantum and classical walks [5]; noise-assisted transport is characterised in the photosynthetic literature [6, 7]; complex edge phases yield directed walks [9]. Most directly, unitary CTQW centrality was recently applied to residue interaction networks with an IBM hardware demonstration [2].

The gap. Existing quantum work on residue networks is centrality-based on purely topological graphs: it asks which residues are structurally important, not which are *specifically coupled to the active site*, and leaves the coherence assumption unexamined at physiological temperature. We differ in five respects: (i) evolutionary conservation enters as on-site energy on the diagonal, adding functional prior without densifying H or breaking Hermiticity; (ii) an optional chiral phase from elastic-network mode direction breaks time-reversal symmetry, giving a *directed* connectivity matrix no symmetric classical measure can express; (iii) coarse-graining is anchored on the active site by a Krylov projection preserving source transport exactly; (iv) the deliverable is a deterministic, training-free spectral object; and (v) the coherent fraction is swept, not fixed.

3. Technical Approach

3.1 Hamiltonian encoding

Nodes are residues, which keeps the output at the residue resolution the deliverable requires. H is sparse and Hermitian:

$$H_{ij} = w_{ij} e^{i\phi_{ij}} \quad (i \neq j), \quad H_{ii} = c_i. \quad (1)$$

Three physically distinct channels are fused, each entering a different part of H so that none destroys the sparsity or the Hermiticity of the others.

Off-diagonal magnitude w_{ij} — mechanical topology. Nodes are residues represented by $C\alpha$; an edge exists where the $C\alpha$ – $C\alpha$ distance falls within 7.3 Å, optionally weighted by heavy-atom contact counts at 4.5 Å. This is the contact network underlying the Gaussian Network Model, whose Kirchhoff matrix is $\Gamma = D - A$. We borrow the *contact graph* only, with no GNM mode decomposition: the dynamics is generated by H , not by Γ ’s slow modes. The graph is naturally sparse (degree 6–10), which bounds circuit cost downstream. Geometry beyond connectivity enters through the phase channel rather than as a separate distance matrix, which would be largely redundant with the mechanical contacts.

Diagonal c_i — evolutionary prior. Per-residue conservation from ESM-2 enters as on-site energy. Carrying the functional prior on the diagonal injects sequence information without adding edges, so H stays sparse and Hermitian and the circuit cost is unchanged. This channel separates the method from purely topological prior work, and is accompanied by an ablation against the topology-only configuration.

Phase ϕ_{ij} — directionality. The projection of the lowest anisotropic-network-model mode along edge (i, j) [11] supplies a complex phase; ANM is an analytic eigenproblem of the elastic Hessian, not a sampling procedure, so this stays within the no-MD constraint. This channel is central, not decorative: the directed readout of Section 3.2 vanishes without it. Channels are z-scored before fusion and asymmetric inputs symmetrised. The walk is generated in Laplacian form (off-diagonal $-H_{ij}$ with degree on the diagonal), which the ablation selects as giving the strongest directed signal; the topology-plus-phase configuration is the working model, and the conservation diagonal is carried as an ablated option rather than a default.

Hermiticity holds channel by channel: c_i real gives $H_{ii} = \overline{H_{ii}}$; contacts are symmetrised as $(A + A^T)/2$; antisymmetric phases $\phi_{ji} = -\phi_{ij}$ give $H_{ji} = \overline{H_{ij}}$. With phases H is complex Hermitian but non-symmetric: still a valid walk, yet capable of directed connectivity [9]. Since only phase around cycles is physical, we gauge-fix and assert $\|H - H^\dagger\|_F < 10^{-10}$ as a pre-flight check. All inputs derive from apo coordinates and sequence, and the anisotropic modes are an analytic eigenproblem rather than MD sampling, so the no-MD constraint is satisfied.

3.2 Readout: the directed connectivity matrix

The propagator is $U(t) = e^{-iHt}$. The deliverable is its net *directed* current at the natural time τ fixed by the spectral bandwidth,

$$J_{ij}(\tau) = |U_{ij}(\tau)|^2 - |U_{ji}(\tau)|^2, \quad (2)$$

the $N \times N$ directed connectivity matrix the challenge requires. Its defining property is diagnostic: J is *identically zero* for real-symmetric (achiral) H , so a non-zero directed current exists only when the chiral phase breaks time-reversal symmetry. It is therefore a genuinely quantum observable that no symmetric classical measure — communicability e^A [8], diffusion e^{-Lt} , or the average mixing matrix [1] — can express on the same graph. To remove the single-time choice while remaining coherent, we deliver the time-averaged coherent current of the quantum stochastic walk at full coherence ($p = 0$); communicability and average mixing are retained only as classical comparators. The reason this readout, and not a symmetric one, is the right object is structural: a proximity-loaded symmetric connectivity ranks hubs and near neighbours, whereas allosteric residues are distal, so the directed, coherence-borne current is what separates them from background.

3.3 Ranking metric

Raw connectivity is confounded by degree and by proximity: hubs and residues near S score highly for reasons unrelated to allostery. The ranking metric must express *specific* directed coupling to the active site S relative to background:

$$s(j) = \sum_{i \in S} J_{ij}, \quad \rho(j) = \frac{s(j)}{\sum_k |J_{kj}|}, \quad z(j) = \frac{\rho(j) - \mu_{\text{bg}}}{\sigma_{\text{bg}}}, \quad (3)$$

with $\mu_{\text{bg}}, \sigma_{\text{bg}}$ from background residues (non-active, non-neighbour surface residues and non-functional pockets). Because allosteric sites are distal by definition, the delivered score removes the geometric confound at source: $z(j)$ is standardised within $C\alpha$ -distance shells from S , so the proximity component every transport carries is subtracted and what remains is coupling *beyond* geometry. This distance-deconfounded $z(j)$ is the delivered ranking, and is by construction the same quantity the challenge’s primary objective tests, removing any mismatch between what we optimise and what we are scored on.

Hit selection. We exclude S and its first neighbours, enforce a distal constraint ($C\alpha-C\alpha > 12 \text{ \AA}$ from S), then take the five highest $z(j)$, merging spatially adjacent residues into one hit. Pocket-ness is not used as a filter: cryptic pockets are closed in apo structures and fpocket will miss them, so a hard druggability gate would defeat the purpose of the task. For targets with formed pockets we cluster high- z residues and annotate pocket-ness as a druggability flag; for c-Myc, disordered and without a static pocket, the annotation is off.

3.4 Active-site-anchored dimensionality reduction

Ranking requires only the source-to-all transport vector $\langle j|e^{-iHt}|\psi_S\rangle$, and that dynamics is spanned *exactly* by the Krylov subspace $\mathcal{K}_K(H, \psi_S) = \text{span}\{|\psi_S\rangle, H|\psi_S\rangle, \dots, H^{K-1}|\psi_S\rangle\}$ generated from the normalised source state $|\psi_S\rangle = \sum_{i \in S} a_i |i\rangle$. Lanczos iteration orthogonalises it via

$$\beta_r |q_{r+1}\rangle = H|q_r\rangle - \alpha_r |q_r\rangle - \beta_{r-1} |q_{r-1}\rangle, \quad \alpha_r = \langle q_r | H | q_r \rangle, \quad (4)$$

giving $Q = [q_1 \dots q_K]$ with $q_1 = |\psi_S\rangle$ and a tridiagonal projection $H_K = Q^\dagger H Q$ carrying α_r on the diagonal and β_r off it. Since $Q^\dagger |\psi_S\rangle = e_1$,

$$\langle j | e^{-iHt} |\psi_S\rangle = [Q]_{j,:} e^{-iH_K t} e_1, \quad (5)$$

so source transport is preserved exactly in K dimensions. For the quantity that determines the ranking, retention under coarse-graining is therefore an algebraic identity, not an empirical overlap measure, which is what the challenge asks to be demonstrated. Generic spectral clustering preserves only generic modes and destroys within-cluster residue resolution [4]. We use $K = 32-64$ for analysis and $K = 8-16$ on hardware, decoupling qubit count from protein length. The delivered $N \times N$ directed matrix J is computed by exact matrix exponential at full resolution (single-particle walks live in N dimensions, so this is polynomial and trivial for $N \approx 300-1000$); the Krylov route supplies the hardware track, and agreement between the two is a required consistency check.

3.5 Back-mapping the reduction to the $N \times N$ matrix

The walk runs in K dimensions, whereas the deliverable is the $N \times N$ residue matrix; the Lanczos isometry Q , whose columns already live in residue space, connects them. The directed current is read in the K -dimensional chain and lifted to residues by $|Q|^2$; by the exactness identity (5), the source column that the ranking consumes is reconstructed exactly, error appearing only for residue pairs spectrally distant from the active site. As $K \rightarrow N$ the reduction becomes the full-resolution readout, so the classical deliverable and the hardware-track reconstruction are one estimator at two truncations, not two methods. The per-residue coverage $(QQ^\dagger)_{ii} \leq 1$ is reported as a truncation diagnostic alongside the coarse-graining fidelity of Section 5. On hardware only the K -dimensional evolution is executed; Q is a classical $N \times K$ matrix, so back-mapping is post-processing and adds nothing to the qubit budget.

3.6 Coherence sweep

Interference here is a modelling choice, not a claim about physiological coherence. Embedding H in a quantum stochastic walk [5] weights coherent and incoherent channels by a single parameter $p \in [0, 1]$:

$$\frac{d\rho}{dt} = -i(1-p)[H, \rho] + p \sum_{i,j} \left(L_{ij} \rho L_{ij}^\dagger - \frac{1}{2} \{L_{ij}^\dagger L_{ij}, \rho\} \right), \quad (6)$$

with jump operators $L_{ij} = \sqrt{\gamma_{ij}} |i\rangle\langle j|$ acting only along existing edges, so both channels share the same topology. At $p \rightarrow 0$ only the von Neumann term survives and the dynamics is the coherent walk e^{-iHt} ; at $p \rightarrow 1$ the populations obey the classical master equation with generator L , giving e^{-Lt} . The sweep is the *ablation* and the *noise-resilience study* at once, since the classical control is our own endpoint on an identical graph and generator, not a separate model. Its outcome is also the central evidence: the directed current is zero at the incoherent endpoint, so the ranking decays from its coherent value to chance as $p \rightarrow 1$. Environment-assisted transport predicts an interior optimum [7]; we test for one rather than assume it, and finding the coherent endpoint optimal is itself evidence that the signal is carried by coherence, not a failed search.

3.7 Learning components and quantum advantage

The predictor is training-free, its principal robustness argument. Learning appears only where it cannot leak: ESM-2 supplies the conservation diagonal, ablated against the topology-only configuration; Bayesian optimisation sets the two to four thresholds by leave-one-protein-out on a disjoint calibration set; and a language-model layer turns the ranked output into a structured brief, so chemists can use the tool without quantum training.

Single-particle walks are classically polynomial, and we claim no speed-up: predictions come from exact classical simulation, which the challenge permits as a quantum-inspired approach. The quantum content is the physics of the model, coherent interference evidenced by discriminative gain over e^A , plus a credible near-term hardware path. Genuine hardware advantage arises only for multiple simultaneous excitations under exchange coupling, where the state space grows as $\binom{N}{m}$ ($\binom{100}{3} = 161,700$, $\binom{100}{5} = 75,287,520$), beyond open-system classical simulation; that regime is a Phase II stretch goal.

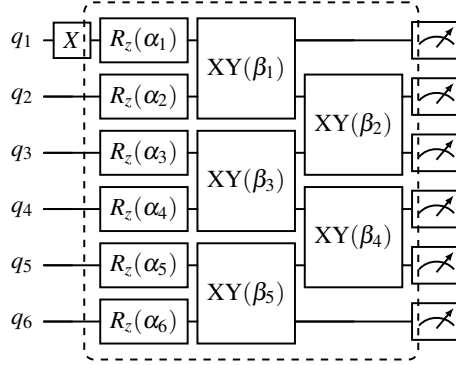


Figure 1: The propagation circuit at $K = 6$; production runs use $K = 8\text{--}16$ with identical structure. The dashed block is one Trotter step, repeated r times. State preparation is the single X gate, since $|\psi_S\rangle$ is e_1 in the Krylov basis; each step applies the diagonal α_r as parallel R_z rotations, then the off-diagonal β_r as two layers of $XY(\beta) = \exp[-i\beta(XX + YY)/2]$ on odd and even bonds, each compiling to two CNOTs plus single-qubit rotations. Measurement is in the computational basis; shots outside Hamming weight one are discarded.

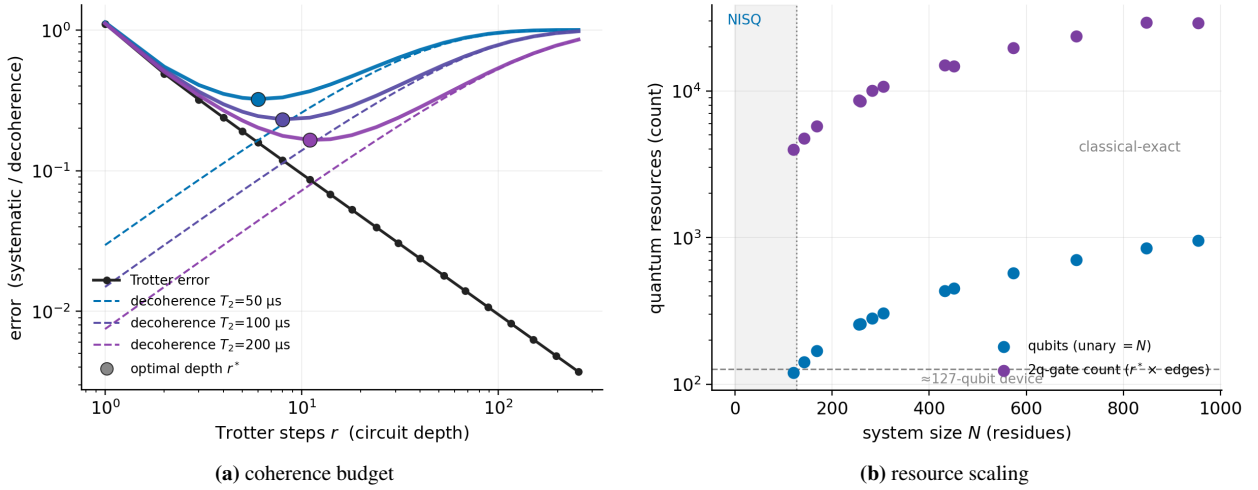


Figure 2: The coherence-time limitation, quantified on the current model. (a) Systematic Trotter error falls with circuit depth r while decoherence infidelity rises; their sum is minimised at an optimal depth r^* that grows as $\sqrt{T_2}$. The ranking converges well inside r^* , so a hardware demonstration sits within the coherence budget. (b) Unary qubit count equals residue count and two-qubit gate count scales with contact edges: small targets are demonstrable on near-term devices, larger ones are delivered on the exact classical track.

3.8 Circuit specification and resource estimate

H_K is tridiagonal, hence a one-dimensional open chain with on-site energies α_r and nearest-neighbour hoppings β_r . The Krylov step therefore maps an arbitrary protein contact topology onto the structure near-term hardware executes best: no long-range couplings, no SWAP overhead, full parallelisation.

We use a unary encoding: K Krylov basis states map to K qubits, the working space being the Hamming-weight-one sector. Figure 1 gives the resulting circuit. Because the chain is one-dimensional, two odd/even layers exhaust every bond, so *two-qubit depth is independent of K and set only by the Trotter number r* , which answers the coherence-time constraint directly. Gate counts follow immediately: $2(K-1)r$ two-qubit gates at a two-qubit depth of $4r$.

Reading the coherence budget. Two errors must be separated (Fig. 2a). The systematic Trotter error falls with depth; decoherence rises with it; their sum sets an optimal depth $r^* \propto \sqrt{T_2}$. What makes the method feasible is that the *ranking* converges at a depth well below r^* , so the walk is read before it decoheres. The binding constraint is accumulated gate error rather than duration, which fixes the headline run at the smallest K and the smallest Trotter number passing a simulator convergence test and declares the large- K circuits simulator-only. Post-selection on excitation number turns much of the residual error into discarded shots rather than biased estimates, making it the primary mitigation and ZNE the secondary one.

Readout, platform and budget. Sampling a few evolution times gives the source row of J directly, since the excited qubit labels the Krylov basis vector; results project back to residues by Q . Because the chain is one-dimensional, all-to-all connectivity buys nothing: superconducting devices on Braket (IQM, Rigetti) are the technical and economic fit, with per-shot rates well below trapped-ion alternatives, and the time points go as a single program set. Classiq supplies depth-constrained synthesis under a fixed coherence budget, reported before and after optimisation. A full campaign — simulator validation, noise modelling at $K \leq 12$, a headline run with zero-noise extrapolation, a K -scaling study and a cross-platform check — is on the order of a minute of pure QPU time, dominated by queueing and classical processing.

Table 1: Data and tools. No training set is required.

Purpose	Source or tool
Required targets	RCSB PDB: 4OBE→6OIM, 1OPL→5MO4, 5TBY→6C1H, 1NKP (c-Myc)
Calibration set	ASD, AlloBench: deduplicated by UniProt, length-binned, fixed seed; disjoint from targets
Structure and features	BioPython, ProDy (contact graph, ANM modes), fair-esm (ESM-2 conservation)
Simulation and readout	NumPy / SciPy matrix exponential for J ; QuTiP for the Lindblad sweep
Circuits and hardware	Qiskit, AWS Braket SDK, Classiq (synthesis, depth optimisation), Mitiq (ZNE)
Baselines and output	fpocket, NetworkX, ProDy GNM centrality; PyMOL and browser viewer; LLM brief

Table 2: Blind validation targets. The apo structure is the sole input; holo defines ground truth only after predictions are frozen.

Target	Apo	Holo	Site to blind-predict	Ref.
KRAS G12C	4OBE	6OIM	Cryptic Switch-II pocket (sotorasib)	[15]
BCR-ABL1	1OPL	5MO4	Distal myristoyl pocket (asciminib)	[16]
Cardiac myosin	5TBY	6C1H	Site stabilising the super-relaxed state	[17]
c-Myc	1NKP	—	Myc–Max interface; pocket annotation disabled	—

4. Implementation Plan

Phase I delivers this specification; Phase II builds and validates it. The circuit of Section 3.8 is the defining construction; because its single-excitation sector is exactly solvable, the same directed matrix J is computed classically at full resolution, serving as both deliverable and hardware reference, so no milestone depends on device availability.

1. **Structure and features.** Fetch apo structures, select the catalytic domain, strip waters and heteroatoms, build the heavy-atom contact graph, and compute per-residue conservation.
2. **Hamiltonian assembly.** Fuse magnitudes, diagonal and optional phases with z-score normalisation; gauge-fix; assert Hermiticity.
3. **Readout.** Exact matrix exponential to the directed current J , delivered as its time-averaged coherent form.
4. **Ranking.** Compute s , ρ and z ; apply distal and neighbour filters; return the top five residues with optional pocket annotation.
5. **Coherence sweep.** Lindblad dephasing scan from $p = 0$ to $p = 1$, reported against e^A and e^{-L_t} .
6. **Validation.** Blind runs on the four targets, then extended generalisation runs, against all baselines in Section 5.
7. **Hardware.** Execute the circuits of Section 3.8 on Braket, verifying the quantum J against the classical one under post-selection and zero-noise extrapolation.

Data processing and calibration set. Structures are reduced to a single chain, stripped of waters and heteroatoms, and mapped to a $C\alpha$ contact graph; UniProt active-site annotations are aligned to the resolved sequence. Ground truth is frozen by the challenge rule — residues with a heavy atom within 4.5 Å of an allosteric-ligand heavy atom in the superposed holo structure. The calibration set is drawn from the pooled ASD and AlloBench allosteric entries, deduplicated by UniProt and stratified into length bins under a fixed seed, giving ten proteins that span the size range and are disjoint from the blind targets; thresholds are fixed on it by leave-one-protein-out. No labels enter the predictor, so the set calibrates but does not train.

Deliverables. (i) The $N \times N$ directed connectivity matrix J at full residue resolution. (ii) Ranked top-five allosteric residues per target by $z(j)$. (iii) This methodological report, identifying the directed current as the quantum metric and its vanishing in the classical limit as its link to biological signal transmission. (iv) An interactive 3D connectivity map showing the protein coloured by $z(j)$ with predicted site and propagation paths, in PyMOL and a browser viewer. The map, not the matrix, is the artefact a chemist acts on, and the language-model brief accompanies it.

5. Validation Plan

Ground truth and controls. Positive labels come from the holo structure and are used only after predictions are frozen: after superposition onto apo coordinates, residues with a heavy atom within 4.5 Å of an allosteric-ligand heavy atom are labelled allosteric, with indices mapped back through the apo–holo alignment. Two negative sets match the controls named in the challenge. Random background draws surface residues excluding the active site, its first neighbours and the labelled site. Non-functional pockets are fpocket pockets on the apo structure overlapping neither the active site nor any annotated functional site, scored by the same $z(j)$. The second is the stricter test, since it excludes the possibility that the method merely separates pocket-like from non-pocket-like surface.

Metrics and statistical tests. Per target we report residue-level ROC-AUC, top-5 and top-10 hit rate, enrichment at top- k , and the minimum $C\alpha$ distance from each predicted residue to the true site centre. Significance is tested against each negative set by Mann–Whitney U and by a permutation test resampling residues matched on solvent accessibility and

distance from the active site, so a geometric confound cannot produce a significant result on its own. We report p -values with rank-biserial effect sizes and bootstrap intervals on ROC-AUC. A target passes when both tests reach $p < 0.05$ after Holm correction across targets and the true site appears in the top five.

Baselines and ablations. Baselines run on the identical graph wherever possible: classical communicability e^A [8], classical diffusion e^{-Lt} , GNM centrality, bond-to-bond propensity [3] as the strongest published method in this setting, PocketMiner at ROC-AUC 0.87 [13] as the learned reference, and a random ranking. The two that share our graph and generator are decisive, since they separate the contribution of interference from that of graph structure. Ablations remove one component at a time: the conservation diagonal, the chiral phase, the degree deconfounding in $\rho(j)$, and the background standardisation, with the readout additionally swapped between the directed current and its symmetric classical comparators. Each ablation is reported as a Δ ROC-AUC so the contribution of every channel is quantified.

Robustness. Noise resilience is assessed on the metric, not only on the state. We sweep the dephasing parameter p , simulate gate-error and finite-coherence models on DM1 at $K \leq 12$, and report the Spearman correlation between noisy and noiseless $z(j)$ with the top-5 overlap: what matters is whether the ranking survives, not whether the state does. Coarse-graining fidelity uses the same two measures between full-resolution J and its Krylov reduction across $K = 8, 12, 16, 32, 64$, complementing the exactness identity of (5) with an empirical curve showing where truncation begins to matter.

Blind protocol and reproducibility. No parameters are fitted to data, so leakage is structurally limited. The thresholds come from leave-one-protein-out on a calibration set disjoint from the targets; ground-truth labels are computed in a module the predictor cannot import, enforced by unit test; and sensitivity sweeps over every threshold are published. Code, environment specification, seeds and intermediate matrices are released so each figure regenerates from a clean checkout.

6. Impact, Metrics of Success and Risks

Where the value sits. The tool acts where a target is advanced or abandoned. Screening without knowing where on the surface to aim is the most expensive form of ignorance in early discovery, and shelved targets carry zero option value. A static structure becomes a ranked surface map in minutes instead of months, and being training-free the method applies immediately to targets with no allosteric precedent: the population supervised predictors serve worst.

Metrics of success. Discrimination against the published 0.87 [13]; interference gain over e^A and e^{-Lt} on the same graph; runtime per target; top-5 hit rate and distance to the true site; and generalisation on untouched ASD targets.

Risks. The directed current depends on the chiral phase, so a target whose functional motion is poorly captured by the lowest ANM mode weakens the signal; the phase construction and the time window are therefore reported with sensitivity sweeps. The conservation diagonal is the largest exposure to the ab-initio-from-topology requirement, which is why the working model is topology and phase only, with the diagonal entering as an ablation. The elastic-network assumption is weakest for c-Myc, which is nonetheless delivered as a required target with pocket annotation disabled. Finally, coherence may carry little signal on a given target, in which case the sweep flattens toward chance; that is a controlled outcome on a question the challenge itself raises, and the classical pipeline remains a working deliverable.

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